

Generation Interconnection System Impact Study Report

For Queue Project Q14-0391

State Campus 138 kV

October 20, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0391. The Interconnection Customer proposes a 130.0 MW solar photovoltaic facility interconnecting at the State Campus 138 kV substation in the Pennsylvania area. The request was received on October 13, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 241 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 17 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$5,967,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of KIYA SUBSTATION-STATE CAMPUS 138kV, the MICHIGAN CAL-GRIFFIN 138kV line loads to 153.9% of its 235 MVA rating. Project DFAX: 23.6%.

The MICHIGAN CAL-SAHARA 138kV Line is overloaded to 148.9% of its 240 MVA emergency rating for the loss of the STATE CAMPUS-KIYA 138kV. The project contributes 23.6% DFAX (30.7 MW) to this overload.

BELL-WINTERS-SAHARA SUBSTATION 138kV: 142.8% loading (246 MVA) following the outage of the KIYA-STATE CAMPUS 138kV line; DFAX = 23.6%.

The BELL-WINTERS-GOETHE-MAYHEW 138kV Line is overloaded to 140.7% of its 248 MVA emergency rating for the loss of the STATE CAMPUS-KIYA 138kV. The project contributes 23.6% DFAX (30.7 MW) to this overload.

For the loss of KIYA-STATE CAMPUS 138kV Line, the GOETHE-MAYHEW SUBSTATION-RONCO 138kV loads to 136.3% of its 253 MVA rating. Project DFAX: 23.6%.

For the loss of AMERICAN RIVER-TONOPAH 138kV, the CWEC IOLA 345-138 kV Transformer Bank loads to 130.1% of its 376 MVA rating. Project DFAX: 35.6%.

The MATANZAS-BELRIDGE 1B 138kV Line is overloaded to 122.5% of its 209 MVA emergency rating for the loss of the JON DAY-MATANZAS SUBSTATION 138kV Line. The project contributes 21.7% DFAX (28.3 MW) to this overload.

FIRSTENERGY FT MARTIN POWER STA-RONCO 345kV: 122.4% loading (1,253 MVA) following the outage of the PRIDES CORNER SUBSTATION-AUURN RD 345kV; DFAX = 24.5%.

The PRIDES CORNER SUBSTATION-FIRSTENERGY FT MARTIN POWER STA 345kV is overloaded to 120.9% of its 1,099 MVA emergency rating for the loss of the AUBURN RD-PRIDES CORNER SUBSTATION 345kV. The project contributes 5.9% DFAX (7.7 MW) to this overload.

For the loss of AUBURN ROAD-PRIDES CONER 345kV Line, the CAVENDISH-POTLACH 138kV L loads to 113.0% of its 231 MVA rating. Project DFAX: 8.3%.

For the loss of KIYA SUBSTATION-STATE CAMPUS 138kV Line, the TUCO-MEDORA SUBSTATION 138kV Line loads to 112.9% of its 261 MVA rating. Project DFAX: 23.4%.

The POTLATCH-GANS 138kV is overloaded to 112.4% of its 237 MVA emergency rating for the loss of the PRIDES CORNER-AUBURN ROAD 345kV L. The project contributes 8.3% DFAX (10.8 MW) to this overload.

The GANS-PIT 7 TAP 138kV is overloaded to 111.8% of its 244 MVA emergency rating for the loss of the AUBURN ROAD SUBSTATION-PRIDES CORNER 345kV. The project contributes 8.3% DFAX (10.8 MW) to this overload.

For the loss of ALLEGHENY ENERGY UNITS 3 4 & 5-SOUTHEAST VICKSBURG 500kV, the WOODLAND TAP-SEWRD 345kV loads to 104.6% of its 1,546 MVA rating. Project DFAX: 16.5%.

The EDES TAP 2-W CAMPUS SUBSTATION 345kV is overloaded to 101.2% of its 1,408 MVA emergency rating for the loss of the ORCHARD ROAD-SHAWVILLE STATION 500kV L. The project contributes 6.0% DFAX (7.7 MW) to this overload.

The SEWARD-KEARNY 345kV is overloaded to 101.1% of its 1,192 MVA emergency rating for the loss of the FT GIBSON-ARMSTROG 500kV. The project contributes 9.1% DFAX (11.9 MW) to this overload.

The LUTESVILLE-EDES TAP 2 345kV Line is overloaded to 100.5% of its 1,421 MVA emergency rating for the loss of the SHAWVILLE STA-ORCHARD RD 500kV. The project contributes 5.7% DFAX (7.4 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
MICHIGAN CAL-GRIFFIN 138kV L	Rebuild line with higher-capacity conductor	\$1,959,000	8% (\$167,000)
MICHIGAN CAL-SAHARA 138kV	Rebuild line with higher-capacity conductor	\$4,070,000	8% (\$348,000)
BELL-WINTERS-SAHARA SUBSTATION 138kV	Rebuild line with higher-capacity conductor	\$4,211,000	8% (\$360,000)
BELL-WINTERS-GOETHE-MAYHEW 138kV Line	Rebuild line with higher-capacity conductor	\$3,755,000	8% (\$321,000)
GOETHE-MAYHEW SUBSTATION-RONCO 138kV	Rebuild line with higher-capacity conductor	\$2,378,000	8% (\$203,000)
CWEC IOLA 345-138 kV Transformer Bank	Replace transformer with higher-rated unit	\$16,981,000	9% (\$1,494,000)
MATANZAS-BELRIDGE 1B 138kV Line	Rebuild line with higher-capacity conductor	\$7,490,000	9% (\$661,000)
FIRSTENERGY FT MARTIN POWER STA-RONCO 345kV	Rebuild line with higher-capacity conductor	\$32,419,000	3% (\$842,000)
FIRSTENERGY FT MARTIN POWER STATION-PRIDES CORNER 345kV	Rebuild line with higher-capacity conductor	\$9,002,000	1% (\$72,000)
CAVENDISH-POTLACH 138kV L	Rebuild line with higher-capacity conductor	\$2,066,000	4% (\$85,000)
TUCO-MEDORA SUBSTATION 138kV Line	Rebuild line with higher-capacity conductor	\$748,000	8% (\$59,000)
GANS-POTLATCH 138kV line	Rebuild line with higher-capacity conductor	\$3,883,000	4% (\$160,000)
GANS-PIT 7 TAP 138kV	Rebuild line with higher-capacity conductor	\$6,200,000	4% (\$256,000)
WOODLAND TAP-SEWARD 345kV	Reconductor line	\$11,090,000	2% (\$244,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
EDES TAP 2-W CAMPUS 345kV	Reconductor line	\$7,515,000	2% (\$190,000)
SEWARD-KEARNY 345kV	Reconductor line	\$6,034,000	2% (\$123,000)
EDES TAP 2-LUTESVILLE 345kV line	Reconductor line	\$14,986,000	3% (\$382,000)
	Total allocated to project		\$5,967,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.